

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION 2011**

**MATHEMATICS
EXTENSION 2**

*Time Allowed – 3 Hours
(Plus 5 minutes Reading Time)*

- *All* questions may be attempted
- *All* questions are of equal value
- Department of Education approved calculators and templates are permitted
- In every question, show all necessary working
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

The answers to all questions are to be returned in separate *stapled* bundles clearly labeled Question 1, Question 2, etc. Each question must show your Candidate Number.

Question 1 (15 Marks)**Marks**

(a) Find:

(i) $\int \frac{e^x}{\sqrt{e^{2x}-1}} dx$ 2

(ii) $\int \frac{1}{x^2-5x+6} dx$ 2

(iii) $\int \frac{d\theta}{2+\cos\theta}$ 3

(b) Evaluate: $\int_{-1}^1 \frac{x}{x^2+2x+5} dx$ 4

(c) If $I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x + 2\sin x} dx$ and $J = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\cos x + 2\sin x} dx$,

(i) Show that $2I - J = \ln 2$. 1

(ii) Evaluate $I + 2J$. 1

(iii) Hence, find the exact values of I and J . 2

Question 2 (15 Marks) START A NEW PAGE

(a) Plot neatly on an Argand diagram the points A , B and C corresponding to the complex numbers w , w^2 and $w\bar{w}$ respectively where $w = \sqrt{3} + i$. 3

(b) Let $z = x + iy$ be a complex number satisfying the inequality 4

$$z\bar{z} + (1-2i)z + (1+2i)\bar{z} \leq 4 \quad \text{where } x \text{ and } y \text{ are real.}$$

Sketch the locus of z on an Argand diagram.

(c) (i) Solve the equation for w : 2

$$w^2 = -11 - 60i.$$

Write your answer in the form $w = x + yi$, where x and $y \in \mathbb{R}$

(ii) Hence, or otherwise, solve the equation: 3

$$z^2 - (1+4i)z - (1-17i) = 0$$

(d) Five girls and three boys are seated at random around a circular table. 3
What is the probability that at least two boys are sitting next to each other?

- (a) $ABCD$ is a cyclic quadrilateral. Chords BE and DF bisect $\angle ABC$ and $\angle ADC$ respectively.

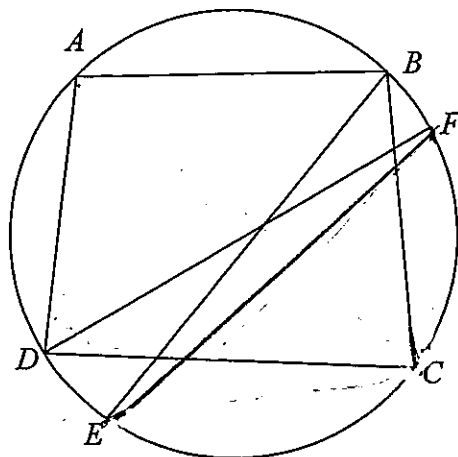


Diagram not to scale

Copy the diagram and prove that EF is a diameter of the circle.

3

- (b) (i) Show whether the function $f(x) = 2|x-1| - |x| + 2|x+1|$ is even, odd or neither, giving reasons. 2
- (ii) Sketch the graph of the function $f(x) = 2|x-1| - |x| + 2|x+1|$, clearly showing all intercepts with the coordinate axes and critical points. Label all branches with the relevant equations. 3
- (c) $P(x_1, y_1)$ is a point on the rectangular hyperbola $xy = 9$.
- (i) Show that the Cartesian equation of the tangent at P is $y_1x + x_1y = 18$. 2
- (ii) Hence, or otherwise, derive the equation of the chord of contact from an external point $T(x_0, y_0)$ to the hyperbola $xy = 9$. 2
- (iii) Prove that the chord of contact is a focal chord when T is a point on the directrix. 3

Question 4 (15 Marks) START A NEW PAGE

Marks

- (a) (i) Find all stationary points for the curve $y^2 = x(3-x)^2$. 3
- (ii) Sketch the curve $y^2 = x(3-x)^2$, showing all stationary points and the intercepts with the coordinate axes. 3

- (b) A particle of mass 2kg is projected vertically upwards with a velocity of $U \text{ ms}^{-1}$ in a medium which exerts a resistive force of $\frac{v}{10}$ Newtons.

- (i) Show that the maximum height H metres reached by the particle is given by: 3

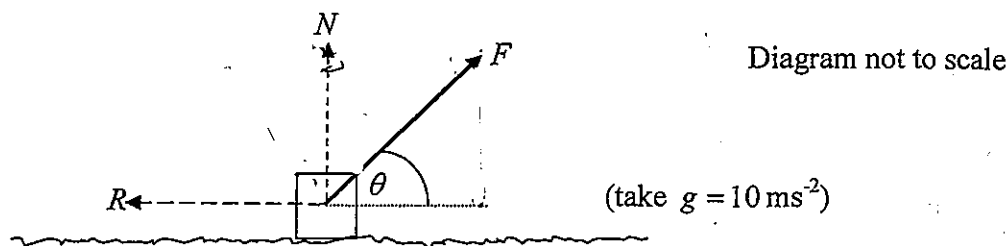
$$H = 20U + 4000 \ln \left(\frac{200}{200+U} \right) \quad (\text{take } g = 10 \text{ ms}^{-2})$$

- (ii) Find the time taken for the particle to reach the maximum height H . 3
- (iii) If $U = 400$, show that the average speed during the ascent is: 3

$$200 \left(\frac{2}{\ln 3} - 1 \right) \text{ ms}^{-1}.$$

Question 5 (15 Marks) START A NEW PAGE

- (a) A block of mass 5 kg is to be moved along a rough horizontal surface by a force (F Newtons) inclined at an angle of θ with the direction of motion where $0 \leq \theta \leq \frac{\pi}{2}$.



The motion is resisted by a frictional force (R Newtons) which is proportional to the normal reaction force (N Newtons) exerted on the block by the surface, such that $R = 0.2N$.

- (i) Show that $F = \frac{50}{5 \cos \theta + \sin \theta}$ Newtons, when the block is about to move. 4
- (ii) Calculate the minimum value of F needed to overcome the frictional resistance between the block and the surface. 2

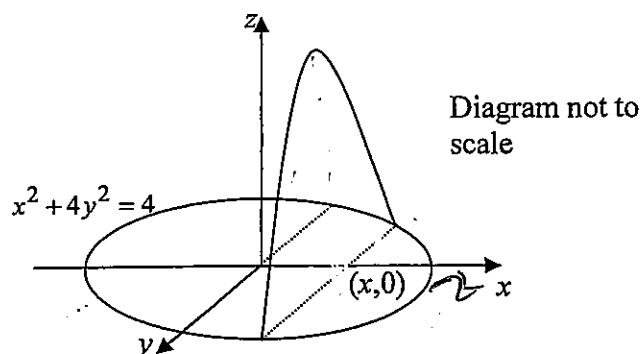
Question 5 continued over page

Question 5 continued

Marks

- (b) (i) A parabola has the equation $x^2 = 4ay$. Show that the area bounded by this parabola and the focal chord perpendicular to the axis is equal to $\frac{8a^2}{3}$ units². 3

- (ii) A solid has an elliptical base whose equation is $x^2 + 4y^2 = 4$ and each cross-section perpendicular to the major axis of the base is a parabola with its focus on the major axis.



- (α) Show that the area of the parabolic cross-section, x units from the origin, is given by the formula

$$A(x) = \frac{4 - x^2}{6} \quad 3$$

- (β) Hence, find the volume of the resultant solid. 3

Question 6 (15 Marks) START A NEW PAGE

- (a) The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ where } \phi > \theta \text{ and } a > b.$$

The points $P'(a \cos \theta, a \sin \theta)$ and $Q'(a \cos \phi, a \sin \phi)$ lie on the auxiliary circle and subtend a right angle at the origin.

- (i) Draw a neat sketch of the above information showing the relative positions of the points P , Q , P' and Q' . 2
- (ii) Express the coordinates of Q in terms of θ . 1
- (iii) The tangents at P and Q meet in point R . Find the coordinates of R in terms of θ . 4
- (iv) Show that R lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$ 1

Question 6 continued over page

Question 6 continued

Marks

- (b) (i) If $\tan(x)\tan(\theta - x) = k$ prove that:

4

$$\frac{1+k}{1-k} = \frac{\cos(2x-\theta)}{\cos\theta}$$

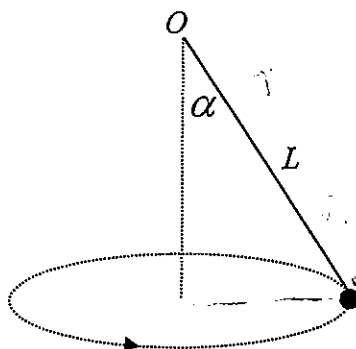
- (ii) Hence, or otherwise, solve the equation for all x .

3

$$\tan x \tan\left(\frac{\pi}{3} - x\right) = 2 + \sqrt{3}$$

Question 7 (15 Marks) START A NEW PAGE

- (a) A particle of mass m kg is fastened to one end of a light inextensible string of length L metres and the other end is attached to a fixed point O . The particle rotates with a uniform angular velocity ω rad/s about a vertical line through O .



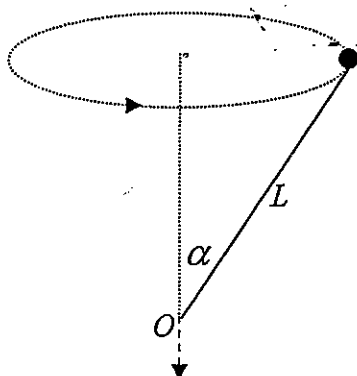
- (i) Show that if α is the angle of inclination of the string to the downward vertical, then $\alpha = \cos^{-1}\left(\frac{g}{L\omega^2}\right)$.

4

- (ii) Explain why steady circular motion is only possible when $\omega^2 > \frac{g}{L}$.

2

- (iii) The point O is now made to descend with a uniform acceleration of $f \text{ ms}^{-2}$, whilst the particle continues to rotate with uniform angular velocity ω .

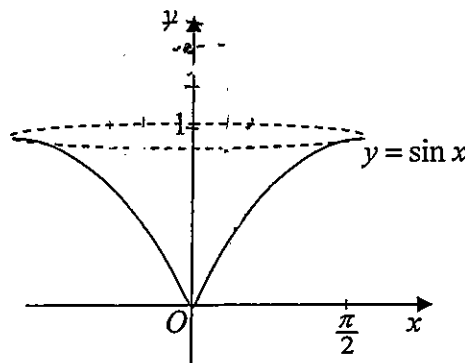


Find f so that the string makes an angle of α with the upward vertical.

3

Question 7 continued over page

- (b) The area between the curve $y = \sin x$, from $x = 0$ to $x = \frac{\pi}{2}$, the y -axis and the line $y = 1$ is rotated about the y -axis.



- (i) Show that the volume of the solid formed can be found by using the formula

3

$$V = \pi \int_0^{\frac{\pi}{2}} x^2 \cos x \, dx$$

- (ii) Hence, calculate the volume of the solid.

3

Question 8 (15 Marks) START A NEW PAGE

- (a) The total number of different groups with 4 members which can be chosen from a group of n people is five times as many as the total number of different groups with 3 members which can be chosen from a group of $n - 2$ people.

3

Find all possible values of n .

- (b) Prove that $\tan^{-1}(5) + \tan^{-1}(3) + \tan^{-1}\left(\frac{4}{7}\right) = \pi$

4

- (c) A curve, defined by the equation $x^2 + 2xy + y^5 = 4$, has a horizontal tangent at the point $P(X, Y)$.

- (i) Show that X is a root to the equation $x^5 + x^2 + 4 = 0$.

3

- (ii) Show that the value of X is between -2 and -1 .

1

- (iii) With the use of a graph, or otherwise, show that X is the only real root to the equation $x^5 + x^2 + 4 = 0$.

4

End of Examination

Suggested Solutions	Marks	Marker's Comments
a) ${}^nC_4 = 5^{n-2} C_3$ $\frac{n!}{(n-4)!4!} = 5^{n-2} \frac{(n-2)!}{(n-5)!3!}$ $\frac{n(n-1)}{(n-4)4} = 5$ $\frac{n(n-1)}{4} = 20(n-4)$ $n^2 - n = 20n - 80$ $n^2 - 21n + 80 = 0$ $(n-16)(n-5) = 0$ $n = 5 \text{ or } 16$	1	
2) $0 < \tan^{-1} 5 < \frac{\pi}{2}$ $0 < \tan^{-1} 3 < \frac{\pi}{2}$ $\therefore 0 < \tan^{-1} 3 + \tan^{-1} 5 < \pi$ $\tan(\tan^{-1} 3 + \tan^{-1} 5) = \frac{3+5}{1-15} = -\frac{4}{7}$ $\therefore \tan^{-1} 3 + \tan^{-1} 5 = \tan^{-1}\left(-\frac{4}{7}\right) + n\pi$ But, as shown above $0 < \tan^{-1} 3 + \tan^{-1} 5 < \pi$ $\therefore n = 1$ $\tan^{-1} 3 + \tan^{-1} 5 = \tan^{-1}\left(-\frac{4}{7}\right) + \pi$ $\tan^{-1} 3 + \tan^{-1} 5 = -\tan^{-1}\left(\frac{4}{7}\right) + \pi$ $\therefore \tan^{-1} 3 + \tan^{-1} 5 + \tan^{-1}\left(\frac{4}{7}\right) = \pi$	1	Many people proved only that $\tan(\text{LHS}) = \tan(\text{RHS})$ without any mention of restrictions. Maximum of 2 in that case (i.e. 1 mark for general solution & equivalent) 1 mark for earlier restriction
3) Differentiate implicitly: $2x + 2y + 2x dy + 5y^4 dy = 0$ w.r.t. x $\frac{dy}{dx}(2x + 5y^4) = -2x - 2y$ $\frac{dy}{dx} = \frac{-2(x+y)}{2x+5y^4}$	1/2	

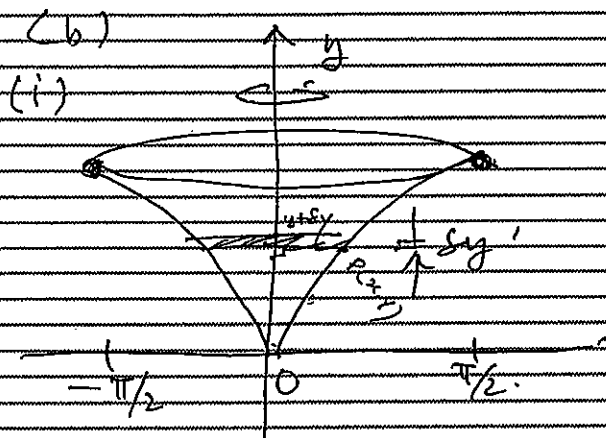
Suggested Solutions	Marks	Marker's Comments
For horizontal tangent $\frac{dy}{dx} = 0$ $\Rightarrow x = -y$ i.e. at P $y = -x$ But P is on the curve so $x^2 + 2xy + y^5 = 4$	1/2	
Substitute $y = -x$ $x^2 - 2x^2 - x^5 = 4$ $x^5 + x^2 + 4 = 0$	1/2	
i.e. x is a root of $x^5 + x^2 + 4 = 0$	1/2	
ii) Let $f(x) = x^5 + x^2 + 4$ $f(-2) = -32 + 4 + 4 = -24$ $f(-1) = -1 + 1 + 4 = 4$	1/2	
f changes sign and since f is continuous between -1 & -2, there must be a root in that domain.	1/2	1/2 reserved for work continuous
iii) $f(x) = 5x^4 + 2x = 0$ at 5 points $x(5x^3 + 2) = 0$ $x = 0$ or $x = \sqrt[3]{-\frac{2}{5}}$ ($x \approx -0.74$)	1	turning points
$f'(x) = 20x^3 + 2$ $f'(0) = 2 > 0$ $\therefore$ Concave up $\therefore$ Local min at $(0, 4)$	1	nature of turning points
$f'\left(\sqrt[3]{-\frac{2}{5}}\right) = -6 < 0$ $\therefore$ Concave down $\therefore$ Local max at $\left(\sqrt[3]{-\frac{2}{5}}, 4.32\right)$	1	graph
As there is no other minimum than that at $x = 0$ and $f(x) \geq 0$, there can be no further roots for $x > 0$.	1	conclusion mentioning only one minimum with that value positive.
As there are no other turning points, the only root is that between $x = -2$ and $x = -1$ so this must be x.	1	

MATHEMATICS Extension 2: Question... 7. (b)

Suggested Solutions

Marks

Marker's Comments



$$\delta V = \pi x^2 \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \pi \sum_0^1 x^2 \delta y$$

$$\therefore V = \pi \int_0^1 x^2 dy$$

$$y = \sin x, \quad \frac{dy}{dx} = \cos x$$

$$\therefore V = \pi \int_0^{\pi/2} x^2 \cos x dx$$

(ii)

$$V = \pi \int_0^{\pi/2} x^2 \frac{d}{dx}(\sin x dx)$$

$$= \pi [x^2 \sin x]_0^{\pi/2} - 2 \int_0^{\pi/2} x \sin x dx$$

$$= 2 \int_0^{\pi/2} x \frac{d}{dx}(-\cos x) dx$$

$$= [-2x \cos x]_0^{\pi/2} + 2 \int_0^{\pi/2} \cos x dx$$

$$\therefore V = \pi \left[\frac{\pi^2}{4} - 2 \right]$$

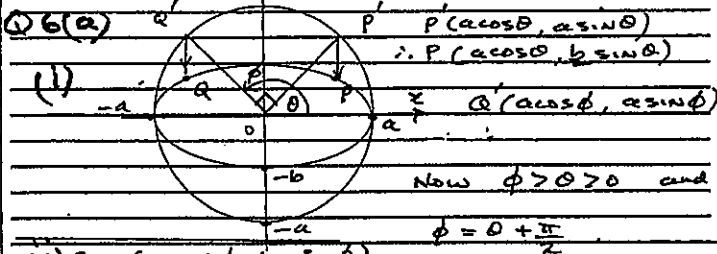
[3 mar

MATHEMATICS Extension 2: Question 6

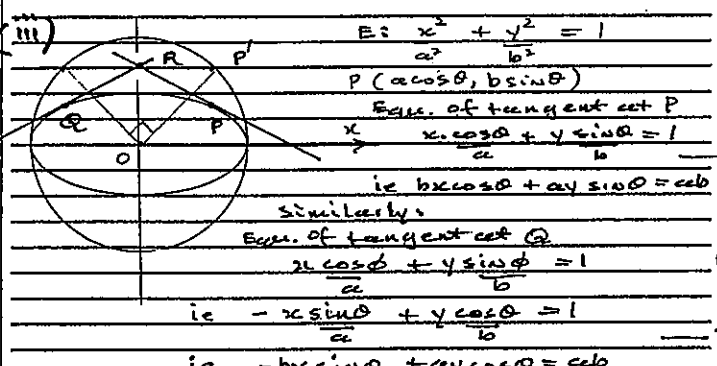
Suggested Solutions

Marks

Marker's Comments



(ii) $Q = (a \cos \phi, b \sin \phi)$
 $= (a \cos(\theta + \frac{\pi}{2}), b \sin(\theta + \frac{\pi}{2}))$
 $Q = (-a \sin \theta, b \cos \theta)$



Solving simultaneously
 (1) $x \sin \theta - b x \sin \theta \cos \theta + a y \sin^2 \theta = ab \sin \theta$ --- (1a)
 (2) $x \cos \theta - b x \sin \theta \cos \theta + a y \cos^2 \theta = ab \cos \theta$ --- (2a)
 Add
 $0 \quad a y \quad = ab(\cos \theta + \sin \theta)$
 $\therefore y = b(\cos \theta + \sin \theta) = b \sqrt{2} \cos(\theta - \frac{\pi}{4})$
 subst in (1) $b x \cos \theta + ab(\cos \theta + \sin \theta) \sin \theta = ab$
 $\Rightarrow x = a(\cos \theta - \sin \theta) = a \sqrt{2} \cos(\theta + \frac{\pi}{4})$
 $\therefore R = (a(\cos \theta - \sin \theta), b(\cos \theta + \sin \theta))$

(iv) LHS $= \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{a^2(\cos \theta - \sin \theta)^2}{a^2} + \frac{b^2(\cos \theta + \sin \theta)^2}{b^2}$
 $= c^2 + s^2 - 2cs + c^2 + s^2 + 2cs$
 $= 2(c^2 + s^2)$
 $= 2 \times 1 = 2$
 $= RHS$

$\therefore R$ is a parametric point on $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$. qed.

$\frac{1}{2}$ For each correct P, Q with $\angle POQ = \frac{\pi}{2}$ [2]

$\frac{1}{2}$ For each correct ordinate. [1]

1 For getting Eqn. of tangent at P
 $M_T = -\frac{b \cos \theta}{a \sin \theta}$

1 For tangent at Q

$\frac{1}{2}$ For correct method that leads to result

1 For getting $y_R = \frac{a}{b} \cos(\theta - \frac{\pi}{4})$ or $x_R = \frac{a}{b} \cos(\theta + \frac{\pi}{4})$ [4]

$\frac{1}{2}$ For $R = \dots$

$\frac{1}{2}$ For a subst. LHS to test

$\frac{1}{2}$ For getting 2 correctly [1]

MATHEMATICS Extension 2: Question 6

Suggested Solutions

Marks

Marker's Comments

(i) LHS: $1+k = 1 + \frac{\tan x + \tan(\theta-x)}{1 - \tan x \tan(\theta-x)}$
 $= \frac{\cos x \cos(\theta-x) + \sin x \sin(\theta-x)}{\cos x \cos(\theta-x) - \sin x \sin(\theta-x)}$
 $= \frac{\cos[x - (\theta-x)]}{\cos[x + (\theta-x)]}$
 $= \frac{\cos(2x-\theta)}{\cos \theta}$
 $\therefore 1+k = \frac{\cos(2x-\theta)}{\cos \theta}$

APPROACH II: $1+k = 1 + \frac{\tan x (\tan \theta - \tan x)}{1 - \tan x (\tan \theta - \tan x)}$
 $= \frac{1 + \tan \theta \tan x}{1 - \tan \theta \tan x}$
 $= \frac{1 + \tan \theta \tan x + \tan x \tan \theta - \tan^2 x}{1 + \tan \theta \tan x - \tan x \tan \theta + \tan^2 x}$
 $= \frac{1 + 2 \tan \theta \tan x - \tan^2 x}{1 + \tan^2 x}$
 $= \frac{1 + 2 \tan \theta \tan x - (\sec^2 x - 1)}{\sec^2 x}$
 $= \cos^2 x [2 - \sec^2 x + 2 \tan \theta \sin x \cos x]$
 $= 2 \cos^2 x - 1 + 2 \sin x \cos x \sin \theta \cos \theta$
 $= \cos 2x + \sin 2x \sin \theta \cos \theta$
 $= \frac{\cos 2x \cos \theta + \sin 2x \sin \theta}{\cos \theta} = \frac{\cos(2x-\theta)}{\cos \theta}$

(ii) $\theta = \frac{\pi}{3}$ so $\frac{\cos(2x-\frac{\pi}{3})}{\cos \frac{\pi}{3}} = \frac{1+2+\sqrt{3}}{1-(2+\sqrt{3})} = \frac{3+\sqrt{3}}{-1-\sqrt{3}}$
 $R = 2+\sqrt{3}$
 $\therefore \cos(2x-\frac{\pi}{3}) = -\frac{(3+\sqrt{3})}{1+\sqrt{3}} = -\frac{(3+\sqrt{3})(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{-3(\sqrt{3}+1)}{\sqrt{3}+1} = -\sqrt{3}$
 $\cos(2x-\frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$
 $\therefore 2x-\frac{\pi}{3} = 2m\pi \pm (\frac{5\pi}{6}), m \in \mathbb{Z}$

$x = \begin{cases} m\pi + \frac{7\pi}{12} \\ m\pi - \frac{\pi}{4} \end{cases} \quad (12m+7)\frac{\pi}{12} \quad (4m-1)\frac{\pi}{4}$

J:\Maths\Suggested Mk sols template_V3.doc

$2x = 2m\pi \pm \frac{5\pi}{6} + \frac{\pi}{3}$
 $x = m\pi \pm \frac{5\pi}{12} + \frac{\pi}{6}$

using $\tan A = \frac{\sin A}{\cos A}$

using $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

(4)

$\frac{\cos(2x-\theta)}{\cos \theta}$ qed.

1 For subst + simplifying

$\frac{-3(\sqrt{3}+1)}{\sqrt{3}+1} = -\sqrt{3}$
 $\frac{1}{\sqrt{3}+1}$

1 For getting 2

1 SOLUTIONS [3]

$E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$. qed.

MATHEMATICS Extension 2: Question 5a

Suggested Solutions

Marks

Marker's Comments

i) Vert. $N + F \sin \theta = mg$
 $\therefore N + F \sin \theta = 5 \times 10$ (1)
 Horiz. $R - F \cos \theta = 0$
 $0.2N = F \cos \theta$
 $N = 5 F \cos \theta$ (2)
 (1) + (2) $5 F \cos \theta + F \sin \theta = 50$
 $F(\cos \theta + \sin \theta) = 50$
 $F = \frac{50}{\cos \theta + \sin \theta}$ #

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

must show
 where 50
 comes from. $\frac{1}{2}m$.
 This is SHOW
 question.

3

ii) F is min when $5 \cos \theta + \sin \theta$ is max
 $5 \cos \theta + \sin \theta = R \sin(\theta + \alpha)$
 $= R \sin \theta \cos \alpha + R \cos \theta \sin \alpha$
 $\therefore R \sin \alpha = 1$, $R \cos \alpha = 5$
 $\therefore R = \sqrt{1^2 + 5^2} = \sqrt{26}$
 $\therefore 5 \cos \theta + \sin \theta = \sqrt{26} \sin(\theta + \alpha)$
 $|\sin(\theta + \alpha)| \leq 1$
 $\therefore \max(5 \cos \theta + \sin \theta) = \sqrt{26}$
 $\therefore \min F = \frac{50}{\frac{5 \times 5}{\sqrt{26}} + \frac{1}{\sqrt{26}}} = \frac{50}{\frac{26}{\sqrt{26}}} = \frac{50}{\sqrt{26}} N$

1

1

1

Pretty well done.

many forget
 max $\sin(\theta + \alpha) = 1$
 $\frac{1}{2}m$

3

Altern.

$F' = \frac{-50}{(5 \sin \theta + \cos \theta)} (-5 \sin \theta + \cos \theta) = 0$
 $\therefore 5 \sin \theta = \cos \theta \therefore \tan \theta = \frac{1}{5}$
 $\therefore \sin \theta = \frac{1}{\sqrt{26}}$, $\cos \theta = \frac{5}{\sqrt{26}}$ ($0 < \theta < 90^\circ$)
 Justify min F
 $\min F = \frac{50}{\frac{5 \times 5}{\sqrt{26}} + \frac{1}{\sqrt{26}}} = \frac{50}{\sqrt{26}} N$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

many forget to
 justify min/max
 $\frac{1}{2}m$.
 many forget $0 < \theta < 90^\circ$
 $\frac{1}{2}m$

MATHEMATICS Extension 2: Question 5b

Suggested Solutions

Marks

Marker's Comments

i) $4ay = x^2 \therefore y = \frac{x^2}{4a}$
 When $y = a$ $4a = \frac{x^2}{a} \therefore x = \pm 2a$
 $A = \int_{-2a}^{2a} (a - y) dx = 2 \times \int_0^{2a} (a - \frac{x^2}{4a}) dx$ (even function)
 $= 2 \times \left[ax - \frac{x^3}{12a} \right]_0^{2a} = 2 \times \left[2a^2 - \frac{8a^3}{12a} \right]$
 $= 2 \times \left[2a^2 - \frac{2}{3}a^2 \right] = 2 \times \frac{4}{3}a^2 = \frac{8a^2}{3} \text{ unit}^2$

1 m

1 m

1 m

Many forget
 even $\frac{1}{2}m$

$\frac{2a}{2} \int \frac{x^2}{4a} dx$ is
 the wrong area
 max $\frac{1}{2}m$

or $x = \pm 2\sqrt{ay}$
 $A = \int_0^a x dy$ (even) $= 2 \int_0^a 2\sqrt{ay} dy$
 $A = 4\sqrt{a} \left[y^{3/2} \cdot \frac{2}{3} \right]_0^a = \frac{8\sqrt{a}}{3} a^{3/2} = \frac{8}{3} a^2 \text{ unit}^2$

1 m

1 m

1 m

many finding
 since answer given

ii) length of latus rectum $= 2y = 4a$
 $a = \frac{y}{2}$
 from (i) $A = \frac{8}{3}a^2 = \frac{8}{3}\left(\frac{y}{2}\right)^2 = \frac{8}{3} \cdot \frac{y^2}{4}$
 $A = \frac{2}{3}y^2$ but $x^2 + 4y^2 = 4$
 $A = \frac{2}{3} \cdot \frac{4 - x^2}{4} = \frac{4 - x^2}{6}$

1 m

1 m

1 m

Some use
 Simpson's Rule
 (must mention)
 $\frac{1}{2}m$
 $\frac{4a}{6} (0 + 4a + 0) = \frac{2}{3} \cdot 4a = \frac{8a}{3}$
 $= \frac{2}{3} \cdot 2y = \frac{2}{3}y^2 = \frac{4 - x^2}{6}$

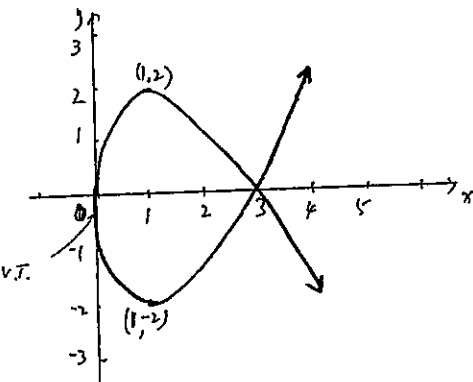
iii) $V = \lim_{\Delta x \rightarrow 0} \sum_{k=1}^n A(x_k) \Delta x$
 $= \int_{-2}^2 \frac{4 - x^2}{6} dx = 2 \int_0^2 \frac{4 - x^2}{6} dx = \frac{2}{6} \left[4x - \frac{x^3}{3} \right]_0^2$
 $= \frac{1}{3} \left(8 - \frac{8}{3} \right) = \frac{16}{9} \text{ unit}^3$ #

$\frac{1}{2}m$

$1 + \frac{1}{2}m$

1

Some forget the
 limit statement $\frac{1}{2}m$

Suggested Solutions	Marks	Marker's Comments
i) $y^2 = x(3-x)^2$ method 1: $y = \pm \sqrt{x(3-x)^2}$ $y' = \frac{3}{2\sqrt{x}} - \frac{3}{2}\sqrt{x} \quad (\text{for } y > 0)$ s.p. $y' = 0$ when $0 = \frac{3}{2\sqrt{x}}(1-x)$ ($x \neq 0$) $\therefore x=1 \quad y=2$ By symmetry s.p. (1,2) or (1,-2) Method 2 implicit differentiation $y^2 = x(3-x)^2 = 9x - 6x^2 + x^3$ $2y y' = 9 - 12x + 3x^2$ s.p. $0 = \frac{3(x-3)(x-1)}{2y} = \frac{3(x-3)(x-1)}{\pm 2\sqrt{x}(3-x)}$ $\therefore x=3 \text{ or } 1 \text{ but } x \neq 3 \text{ as } y=0$ $\therefore x=1 \text{ only}$ s.p. (1,2) or (1,-2)	/ / / / / / / /	FORGOT $\pm$ $-\frac{1}{2}m$ many forgot y values. $-\frac{1}{2}m$ many forgot $\pm$ $-\frac{1}{2}m$ if $x=3$ is included near 2 m. ① Stationary pts ② Vertical tangent at $x=0$ ③ slope at $x=3$ ④ shape, scale, curvature
b) 	3	

Suggested Solutions	Marks	Marker's Comments
4b i) $m\ddot{x} = -mg - \frac{k}{10}$ $\uparrow + \quad \downarrow - \frac{k}{10}$ $2\ddot{x} = -20 - \frac{k}{10}$ $\ddot{x} = -(10 + \frac{k}{20})$ $v \frac{dv}{dx} = -(\frac{v+200}{20})$ $\int dx = \int \frac{-20v}{200+v} dv \quad \text{At max ht } H, v=0$ $H = -20 \int_0^0 1 - \frac{200 dv}{200+v} = 20 \int_0^H 1 - \frac{200 dv}{200+v}$ $H = 20 \left[v - 200 \ln(200+v) \right]_0^H = 20 \left[H - 200 \ln\left(\frac{200+H}{200}\right) \right]$ $H = 20H + 4000 \ln\left(\frac{200}{200+H}\right)$ ii) $\frac{dv}{dt} = -(\frac{200+v}{20})$ $\int dt = \int \frac{-20 dv}{200+v} \quad \text{max ht } H, v=0$ $T = \left[-20 \ln(200+v) \right]_0^H$ $T = -20 \ln\left(\frac{200}{200+H}\right) = +20 \ln\left(\frac{200+H}{200}\right) \text{ sec}$ iii) Ave Speed = $\frac{D}{T} = T =$ $T = 20 \ln\left(\frac{200+400}{200}\right) = 20 \ln 3$ $D = 4000 \times 20 + 4000 \ln\left(\frac{200}{200+400}\right) = 8000 + 4000 \ln\left(\frac{200}{600}\right) = 8000 + 4000 \ln \frac{1}{3}$ $\text{Ave Speed} = \frac{D}{T} = \frac{8000 + 4000 \ln \frac{1}{3}}{20 \ln 3}$ $= \frac{4000(2 + \ln \frac{1}{3})}{20 \ln 3} = 20 \left(\frac{2}{\ln 3} + \frac{\ln \frac{1}{3}}{\ln 3} \right)$ $= 20 \left(\frac{2}{\ln 3} - 1 \right) \text{ m/s}$	1 m $\frac{1}{2}m$ $\frac{1}{2}m$ 1 m 1 m 1 m $\frac{1}{2}m$ $\frac{1}{2}m$ $\frac{1}{2}m + \frac{1}{2}m$	many forgot max ht H, $v=0$ skew equation. must show $-\ln\left(\frac{200+H}{200}\right) = +\ln\left(\frac{200}{200+H}\right)$ did well many forgot unit $-\frac{1}{2}m$ "Shaw" G. must show details. must show $\ln\left(\frac{200+400}{200}\right) = \ln\left(\frac{600}{200}\right) = \ln 3$ $-\frac{1}{2}m$ must show $\ln \frac{1}{3} = -\ln 3$ $-\frac{1}{2}m$ must show factorization $-\frac{1}{2}m$

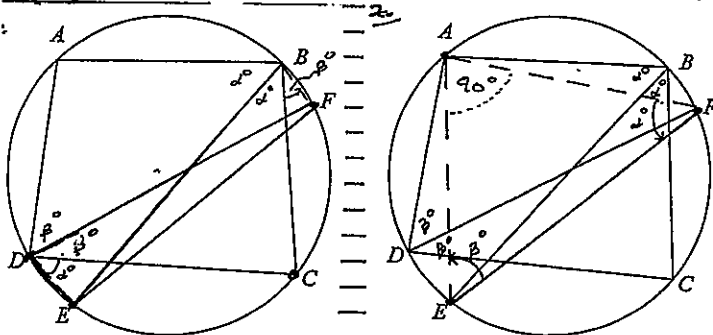
MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

3(a)



Let $\angle ABE = \angle CBE = \alpha^\circ$ $\angle ABC$ is bisected by EB
 $\angle ADF = \angle CDF = \beta^\circ$ $\angle ADC$ " " by FD .

$2\alpha + 2\beta = 180$ (Opposite angles of cyclic quadrilateral $ABCD$ is 180°)

$\therefore \alpha + \beta = 90$

$\therefore \angle EDC = \alpha^\circ$ (Angles subtended on circumference standing on the same arc EC are equal)

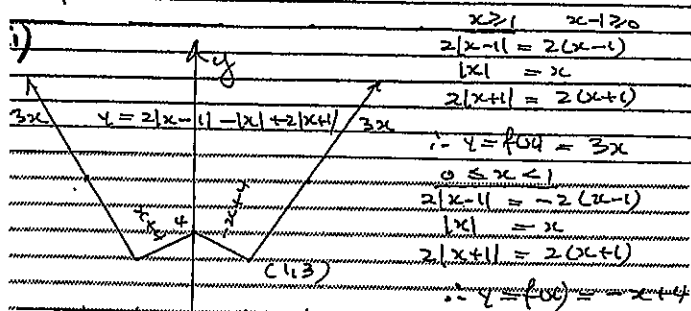
$\therefore \angle EDF = \alpha^\circ + \beta^\circ = 90^\circ$ (adjacent angle addition)

E, D and F exist on circumference EDF
 A right angle at the circumference subtends a diameter
 $\therefore EF$ is a diameter (converse of)

(i) $f(x) = 2|x-1| - |x| + 2|x+1|$

$f(-x) = 2|-x-1| - |-x| + 2|-x+1|$
 $= 2|-1(x+1)| - |-x| + 2|-1(x-1)|$
 $= 2|x+1| - |x| + 2|x-1|$

$\therefore f(-x) = f(x)$
 $\therefore f(x)$ is an EVEN fn



Method 2

1. $\checkmark$
 2. $\angle DAF + \beta^\circ + \alpha^\circ = 180^\circ$
 (angle sum of $\triangle DAF$ is 180°)
 $\therefore \angle DAF = 90^\circ$
 Conclusion - REASON

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

Q3(c) (i)

$xy = 9$
 $\therefore y + x \frac{dy}{dx} = 0$
 i.e. $\frac{dy}{dx} = -\frac{y}{x}$ $\frac{dy}{dx} = -\frac{9}{x^2}$
 $\therefore$ Gradient of tangent at $P(x_1, y_1)$: $m_T = -\frac{y_1}{x_1}$ (or $-\frac{9}{x_1^2}$)
 $\therefore$ Equ. of Tangent at P
 $y - y_1 = -\frac{y_1}{x_1}(x - x_1)$
 $x_1 y - x_1 y_1 = -y_1 x + x_1 y_1$
 $\therefore y_1 x + x_1 y = 2x_1 y_1$, but $x_1 y_1 = 9$ as pt $P(x_1, y_1)$ lies on $xy = 9$
 $\therefore y_1 x + x_1 y = 2 \times 9 = 18$

(ii) Tangent at $P(x_1, y_1)$ is $y_1 x + x_1 y = 18$
 " " $Q(x_2, y_2)$ is $y_2 x + x_2 y = 18$

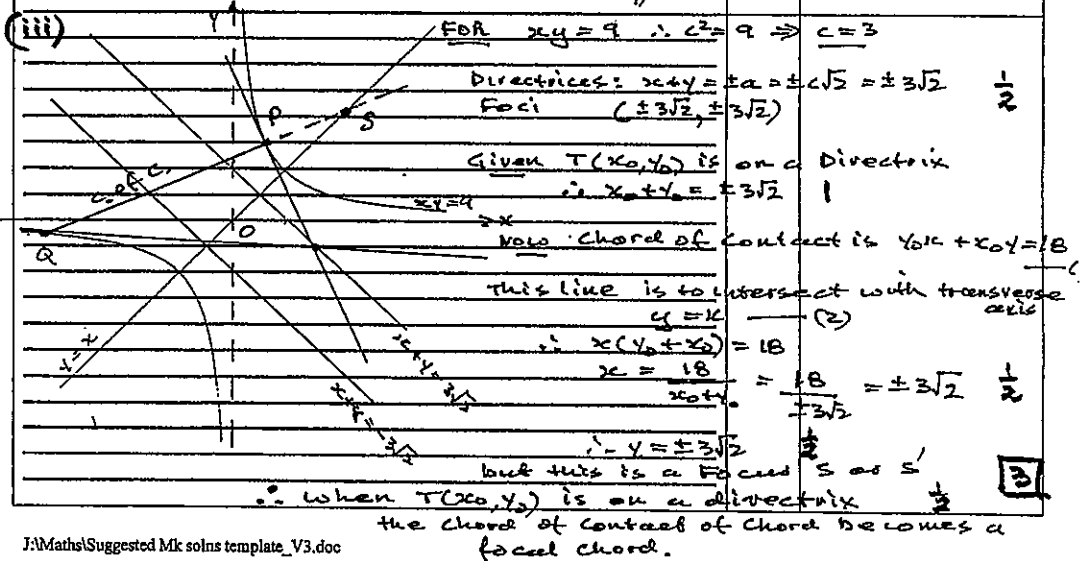
Let PQ be the chord of contact now from $T(x_0, y_0)$

these tangents above pass through $T(x_0, y_0)$

$\therefore y_1 x_0 + x_1 y_0 = y_0 y_1 + x_0 y_1 = 18$ — (1)
 and $y_2 x_0 + x_2 y_0 = y_0 y_2 + x_0 y_2 = 18$ — (2)

So points P and Q lie on the equation $y_0 x + x_0 y = 18$
 which is a line

$\therefore PQ$ uniquely determine the "chord of contact"
 which is of the form $y_0 x + x_0 y = 18$



(d)

(i) Number of ways that 5 girls seated around a circular table = $4!$.

←

(i) or (iii)

1 mark.

(ii) Number of ways that 3 boys seated between the girls.

$$= (5 \times 4 \times 3) \times 4!$$

1st boy 2nd boy 3rd boy

(ii) or (iv)
or equivalent
merit

1 mark.

(iii) 8 people can seat around a circular table $7!$ ways.

←

(iv) No of ways that at least 2 boys are sitting next to each other

$$= 7! - (5 \times 4 \times 3) \times 4!$$

$$= 3600$$

$$\therefore P(E) = \frac{3600}{5040} = \frac{5}{7}$$

1

mark
correct
prob.

MATHEMATICS Extension 2: Question.....(2).

Suggested Solutions

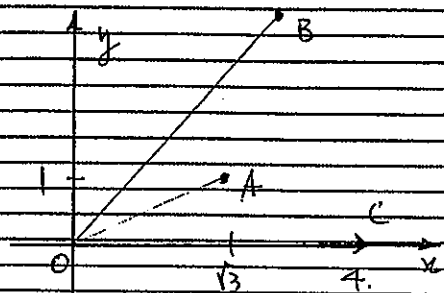
Marks

Marker's Comments

(a) $W = \sqrt{3} + i \left(2 \cos \frac{\pi}{6} \right)$

$W^2 = (\sqrt{3} + i)^2 = 2(1 + i\sqrt{3}) \left(4 \cos \frac{\pi}{3} \right)$

$WW = |W|^2 = 4$



[5].

1 mark
for each
point.

(b) $z\bar{z} = |z|^2 = x^2 + y^2$

Let $z = x + iy$

$(1-2i)(x+iy) = x+iy = 2/x+2y$

$(1+2i)(x-iy) = x-iy + 2ix + 2y$

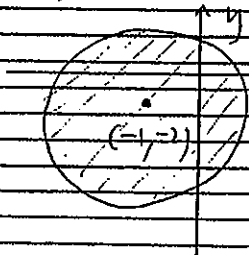
$\therefore z\bar{z} + (1-2i)z + (1+2i)\bar{z} \leq 9$
reduces to

$x^2 + y^2 + 2x + 4y \leq 9$

$(x+1)^2 + (y+2)^2 \leq 9$

Circle centre $(-1, -2)$

$r = 3$



1 mark
substituting
and simplifying
 $z\bar{z}$

1 mark
completing
the square

1 mark
centre, radius
of the circle

1 mark
correct
diagram
for the locus

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

(i) Let $\sqrt{-11-60i} = a+ib$

(ii) $-11-60i = (a^2-b^2) + i2ab$

Equate real and imaginary
parts

$a^2 - b^2 = -11$ $2ab = -60$

$(a^2 - b^2)^2 = (a^2 + b^2)^2 - 4a^2b^2$

$(a^2 + b^2)^2 = 3721$, $a^2 + b^2 = 61$

$2a^2 = 50$ $a^2 = 25$ $a = \pm 5$

$b = \mp 6$

$\therefore W = \pm (5-6i)$

[2]

Correct
quadratic
expression
or
solving a
quadratic equation

Correct
solution
for W.

[3].

Use
quadratic
formula.

Apply (i)
into quad.
formula.

Correct
solution.

(ii) $z^2 - (1+4i)z - (1-17i) = 0$

$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$z = \frac{(1+4i) \pm \sqrt{-11-60i}}{2}$

$= \frac{1+4i \pm (5-6i)}{2}$

$z = 3-i$, $-2+5i$

TRIAL 2011 MATHEMATICS Extension 2: Question...1...

Suggested Solutions

Marks

Marker's Comments

a) Let $I = \int \frac{e^x dx}{\sqrt{e^{2x}-1}}$ Let $u = e^x$
 $"du = e^x dx"$
 $I = \int \frac{du}{\sqrt{u^2-1}}$
 $= \ln|u + \sqrt{u^2-1}| + k$ From tables
 $= \ln(e^x + \sqrt{e^{2x}-1}) + k$
 $(e^x > 0)$

ii) Let $I = \int \frac{dx}{x^2-5x+6}$
 $= \int \frac{dx}{(x-3)(x-2)}$
 $= \int \left(\frac{1}{x-3} - \frac{1}{x-2} \right) dx$
 $= \ln|x-3| - \ln|x-2| + C$
 $= \ln \left| \frac{x-3}{x-2} \right| + C$

iii) Let $I = \int \frac{d\theta}{2+\cos\theta}$ Let $t = \tan \frac{\theta}{2}$
 $"dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta"$
 $"2dt = d\theta"$
 $"Also \cos \theta = \frac{1-t^2}{1+t^2}"$
 $= \int \frac{2dt}{2+2t^2+1-t^2}$
 $= \int \frac{2dt}{t^2+3}$
 $= \frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} + C$
 $= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \tan \frac{\theta}{2} \right) + C$

MATHEMATICS Extension 1: Question...1...(cont)

Suggested Solutions

Marks

Marker's Comments

b) $\int_{-1}^1 \frac{x dx}{x^2+1x+5} = \frac{1}{2} \left\{ \int_{-1}^1 \frac{2x+2-2}{x^2+2x+5} dx \right\}$
 $= \frac{1}{2} \int_{-1}^1 \frac{2x+2}{x^2+2x+5} dx = \int_{-1}^1 \frac{dx}{(x+1)^2+4}$
 $= \frac{1}{2} \left[\ln(x^2+2x+5) \right]_{-1}^1 = \left[\frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) \right]_{-1}^1$
 $= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 4 = \frac{1}{2} \ln 2$
 $= \frac{1}{2} \ln 2 - \frac{\pi}{8}$

c) i) $2I - J = \int_0^{\pi/2} \frac{2\cos x - \sin x}{\cos x + 2\sin x} dx$
 $= \left[\ln(\cos x + 2\sin x) \right]_0^{\pi/2}$
 $= \ln(0+2) - \ln(1) = \ln 2 - 0 = \ln 2$

ii) $I + 2J = \int_0^{\pi/2} \frac{\cos x + 2\sin x}{\cos x + 2\sin x} dx$
 $= \int_0^{\pi/2} dx = \left[x \right]_0^{\pi/2} = \frac{\pi}{2}$

iii) $2I - J = \ln 2$ — (1)
 $I + 2J = \pi/2$ — (2)

2x(2) $2I + 4J = \pi$ — (3)

(1)-(3) $5J = \pi - \ln 2$
 $J = \frac{\pi - \ln 2}{5}$

Substitute into (2)

$I = \frac{\pi}{2} - 2 \left(\frac{\pi - \ln 2}{5} \right) = \frac{5\pi - 4\pi + 4\ln 2}{10}$

$I = \frac{\pi + 4\ln 2}{10}, J = \frac{\pi - \ln 2}{5}$

Generally I knowed
 $\frac{1}{2}$ mark for
 trivial numerical
 errors.
 However in c,iii
 solving simultaneous
 equations requires
 accuracy
 1 or 2 usually.